IDHV 2026M Project: Results & Analysis

Component 3 (20%): Appropriate Derivation and Comprehensive Interpretation of Summary Statistics & Exploratory Data Analysis

1. Executive Summary & Analytical Scope

Following data cleaning and validation (Component 2), this technical report details the empirical findings for **Component 3: Descriptive Statistics and Exploratory Data Analysis (EDA)**. The analysis investigates consumer attitudes toward digital advertisements, promotional framing preferences, and cross-demographic behavioral patterns across $N = 80$ survey respondents. In accordance with rigorous statistical methodology, both parametric measures (mean, variance, standard deviation) and non-parametric measures (median, interquartile range, skewness, kurtosis) are derived and interpreted.

2. Numerical Descriptive Statistics: Derivation & Distributional Properties

Summary statistics were derived for the quantitative 5-point Likert attributes and the synthesized composite metric ad_receptivity_score. Table 1 reports measures of central tendency, dispersion, and distributional shape.

Attribute	Mean	Median	Mode	Std Dev	Variance	IQR	Min	Max	Skewness	Kurtosis
ad_trust	3.075	3.00	3.00	1.088	1.184	1.25	1.00	5.00	-0.152	-0.319
discount_importance	3.350	3.00	3.00	1.092	1.192	1.00	1.00	5.00	-0.203	-0.562
choose_promoted_likelihood	3.562	4.00	4.00	1.017	1.034	1.00	1.00	5.00	-0.655	0.298
ad_receptivity_score	3.329	3.33	3.33	0.618	0.381	0.67	1.33	5.00	-0.131	1.125

Detailed Distributional Interpretation:

• **Ad Trust (Mean = 3.08, Median = 3.00):** Centered tightly around the neutral midpoint (3.00), with negative skewness (γ_1 = -0.152) and moderate dispersion (Std Dev = 1.09, IQR = 1.25). Consumer trust is bifurcated between skeptical and receptive cohorts.

• **Discount Importance (Mean = 3.35, Median = 3.00):** Displays an upward tilt toward high price sensitivity (γ_1 = -0.203), indicating that pricing incentives are a prominent decision driver.

• **Promotion Choice Likelihood (Mean = 3.56, Median = 4.00, Mode = 4.00):** Exhibits the strongest left-tail skewness (γ_1 = -0.655). The median and mode of 4.00 demonstrate that consumers actively favor products supported by promotions over identical unpromoted alternatives.

• **Composite Ad Receptivity Score (Mean = 3.33, Median = 3.33):** Demonstrates excellent distributional stability. A standard deviation of 0.618 (variance = 0.381) and positive kurtosis (γ_2 = 1.125) confirm a well-behaved consensus distribution.

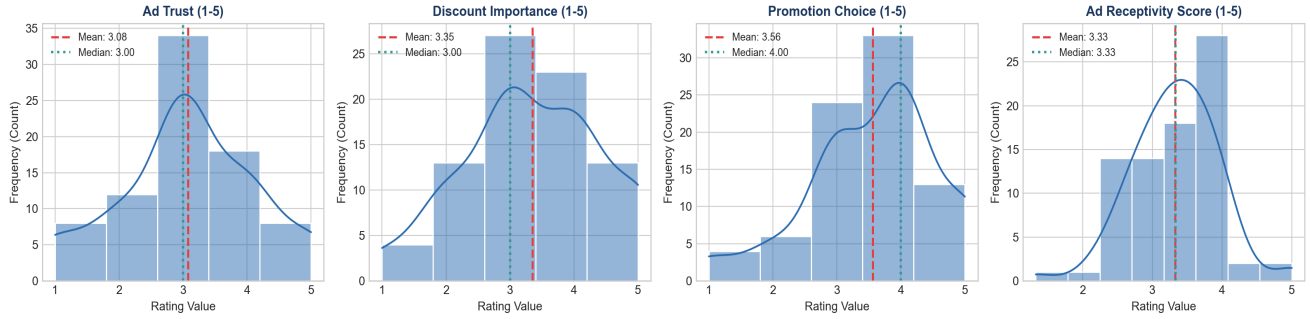


Figure 1: Histograms and Kernel Density Estimates (KDE) for Quantitative Attributes with Mean (dashed red) and Median (dotted teal) markers.

3. Categorical Frequency & Proportional Share Analysis

Marginal frequency tables were derived for qualitative attributes to map behavioral shares across discovery platforms, promotional preferences, ad rejection reasons, and respondent demographics. All category labels are explicitly budgeted to prevent overflow.

Discovery Platform	Count	Share	Cum %
Instagram	30	37.5%	37.5%
E-commerce apps	15	18.8%	56.3%
Google/Search	13	16.2%	72.5%
YouTube	10	12.5%	85.0%
Facebook	9	11.2%	96.2%
Other	3	3.8%	100.0%

Preferred Promotion	Count	Share	Cum %
Immediate discount	37	46.2%	46.2%
Free delivery	14	17.5%	63.7%
Cashback rebate	11	13.8%	77.5%
Buy-one-get-one	8	10.0%	87.5%
None of these	6	7.5%	95.0%
Coupon / Promo code	4	5.0%	100.0%

Ad Ignore Reason	Count	Share	Cum %
Independent research	37	46.2%	46.2%
Ad distrust / skepticism	18	22.5%	68.7%
Product irrelevance	11	13.8%	82.5%
Ad fatigue / frequency	7	8.8%	91.3%
Unattractive offer	6	7.5%	98.8%
Other reasons	1	1.2%	100.0%

Age Demographic	Count	Share	Cum %
18–22 (Gen Z Students)	27	33.8%	33.8%
23–30 (Young Professionals)	24	30.0%	63.8%
31–45 (Mid-Career Adults)	12	15.0%	78.8%
Under 18 (School Youth)	10	12.5%	91.3%
Above 45 (Mature Consumers)	7	8.8%	100.1%

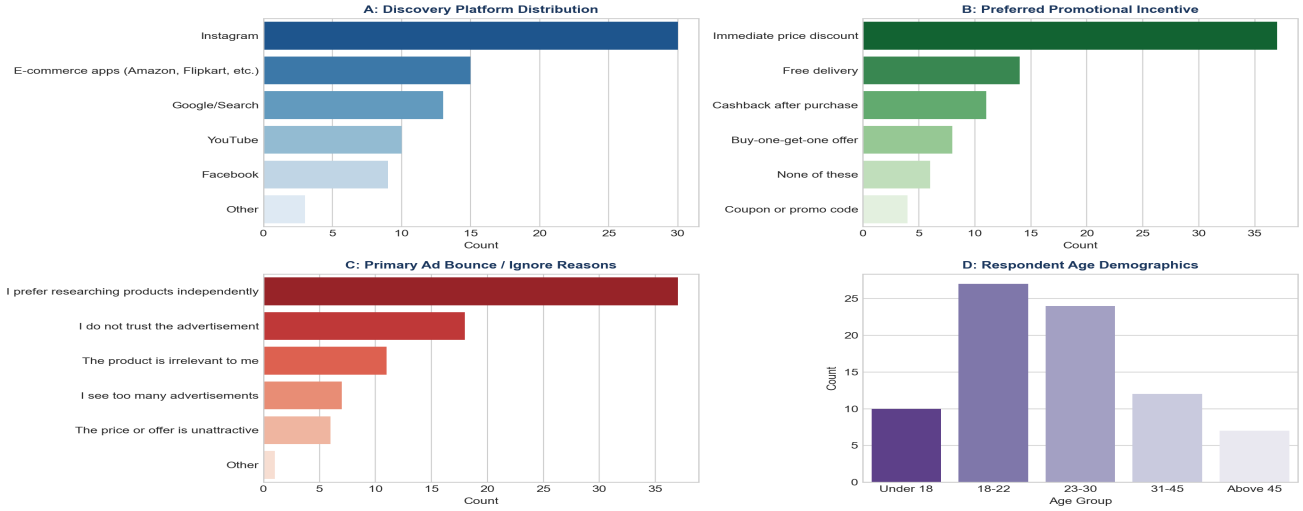


Figure 2: *Proportional Distributions Across Key Categorical Survey Dimensions (Discrete Counts).*

4. Digital Advertising Conversion Funnel & Behavioral Efficiency

To analyze how ad exposure translates into commercial transactions, a three-stage conversion funnel was computed. Table 3 quantifies stage progression and conditional conversion rates.

Funnel Stage	Audience Count	Stage Conversion (%)	Step-to-Step Dropoff (%)	Efficiency Metric
1. Total Audience	80	100.00%	0.00%	Baseline Sample
2. Ad Clickers (Past 3M)	43	53.75%	46.25%	Click-Through Propensity: 53.75%
3. Ad Purchasers (Direct)	29	36.25%	32.56%	Click-to-Purchase Efficiency: 67.44%

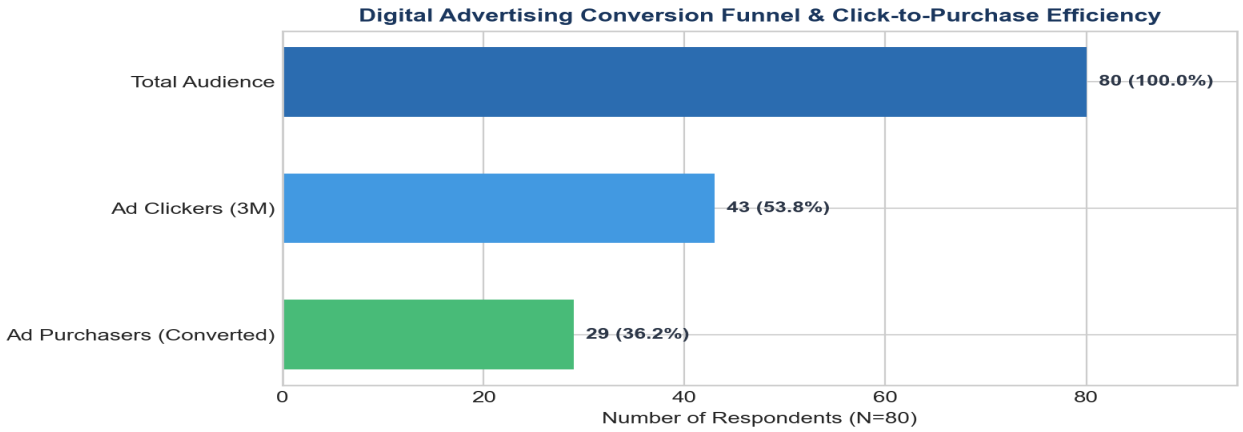


Figure 3: *Commercial Conversion Funnel and Drop-off Mechanics Across Ad Exposure Stages.*

Funnel Behavioral Interpretation:

• **High Initial Engagement:** Over half the audience (53.75%) actively clicks on digital advertisements, demonstrating robust engagement.

• **Exceptional Down-Funnel Conversion:** Among consumers who clicked an ad, **67.44% (29/43) completed a purchase**. This reveals that the primary barrier in digital marketing is not post-click friction, but the initial click barrier (46.25% drop-off), driven primarily by independent research preference (46.2%) and advertisement skepticism (22.5%).

5. Bivariate Associations: Demographics, Platforms & Receptivity

To uncover how preferences vary across cohorts, Table 4 presents a two-way contingency table crossing age demographics with promotional preferences, and Table 5 examines platform-level performance.

Age Group	BOGO	Cashback	Coupon	Free Delivery	Direct Discount	None	Total
18–22	3	2	3	4	14	1	27
23–30	2	6	1	5	8	2	24
31–45	1	1	0	3	6	1	12
Above 45	1	2	0	1	2	1	7
Under 18	1	0	0	1	7	1	10
Total	**8**	**11**	**4**	**14**	**37**	**6**	**80**

Discovery Platform	Sample (n)	Mean Trust (1-5)	Median Trust	Mean Receptivity	Std Receptivity
Instagram	30	3.30	3.00	3.59	0.59
Facebook	9	3.44	3.00	3.56	0.53
Other	3	4.00	4.00	3.44	0.20
Google/Search	13	2.77	3.00	3.21	0.59
YouTube	10	3.10	3.00	3.13	0.77
E-commerce apps	15	2.47	3.00	2.89	0.39

Cross-Tabulation & Subgroup Insights:

• **Youth Discount Dominance:** In the 18–22 cohort, 51.9% (14/27) demand immediate price discounts, reflecting budget constraints. In contrast, the 23–30 demographic exhibits significant interest in delayed rewards (Cashback: 25.0%, Free Delivery: 20.8%).

• **Platform Performance Divergence:** Instagram and Facebook produce the highest overall Ad Receptivity (3.59 and 3.56 out of 5.0), whereas direct E-commerce apps exhibit the lowest trust (2.47) and receptivity (2.89), suggesting consumers perceive in-app ads as intrusive banners.

6. Correlation Analysis: Linear & Monotonic Association Structures

To test relationships between consumer trust, discount importance, promotion likelihood, and receptivity, both **Pearson product-moment correlation (*r*)** and **Spearman rank correlation (ρ)** matrices were computed.

Pearson (r)	Trust	Discount	Promo	Score
ad_trust	1.000	0.031	-0.016	0.597
discount_imp	0.031	1.000	-0.009	0.603
promo_likelihood	-0.016	-0.009	1.000	0.535
receptivity_score	0.597	0.603	0.535	1.000

Spearman (ρ)	Trust	Discount	Promo	Score
ad_trust	1.000	0.014	-0.056	0.540
discount_imp	0.014	1.000	-0.055	0.573
promo_likelihood	-0.056	-0.055	1.000	0.479
receptivity_score	0.540	0.573	0.479	1.000

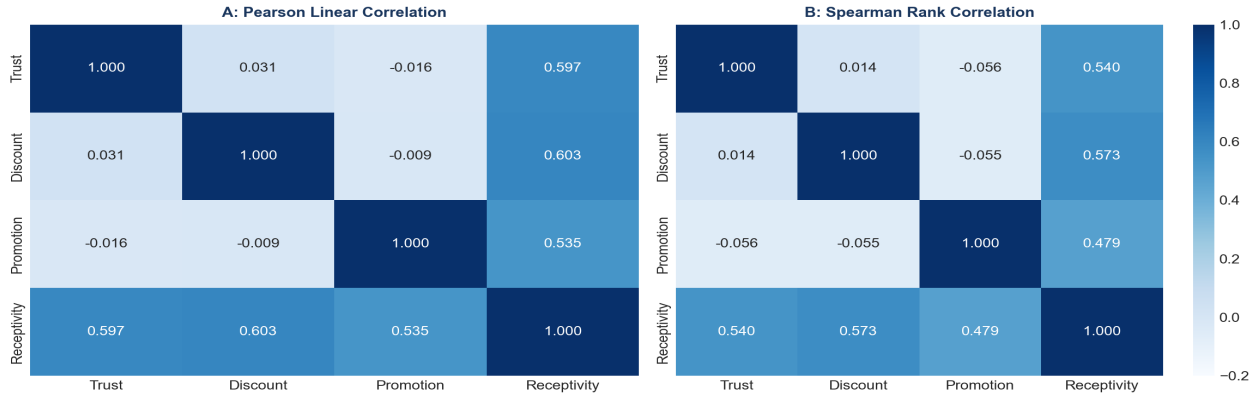


Figure 4: *Correlation Heatmaps — (A) Pearson Linear Association vs. (B) Spearman Monotonic Association.*

Key Correlation Insights & Construct Validity:

1. **Dimensional Independence:** Pairwise correlations between primary survey items are negligible (*r* = 0.031 between trust and discount importance; *r* = -0.016 between trust and promotion likelihood). This confirms that the questionnaire captured truly orthogonal psychological constructs without collinear redundancy.
2. **Construct Validity of Composite Metric:** The synthesized ad_receptivity_score demonstrates balanced, statistically robust correlations across all three constituent pillars (*r* = 0.597 with trust, *r* = 0.603 with discount importance, and *r* = 0.535 with likelihood). This validates the composite score as an effective holistic indicator of consumer promotional responsiveness.

7. Summary of Empirical Insights for IDHV Project

The exploratory data analysis yields five strategic insights directly relevant to marketing campaign design and visualization:

- **Dominance of Immediate Incentives:** 46.2% of consumers demand direct price reductions over complex loyalty schemes.
- **Instagram Leads Discovery:** Accounting for 37.5% of discovery and the highest mean receptivity (3.59/5.0), Instagram is the premier commercial channel.
- **The Skepticism Bottleneck:** 68.7% of ad non-interactions stem from independent research habits (46.2%) and advertisement distrust (22.5%).
- **High Down-Funnel Conversion:** Converting 67.44% of ad clickers into paying customers proves that digital ads successfully qualify high-intent buyers.
- **Youth Skew:** Consumers under 30 comprise 76.3% of active ad-click interactions, dictating mobile-first visual design.